

What Makes Regions Prosper?

Skill Complexity and Development in Europe

1. Research problem

Understanding why some regions thrive while others stagnate is especially relevant in the European Union, where regional inequalities challenge the goals of economic and social cohesion, and here, in Hungary, where economic development is highly uneven across regions.

In this proposal, we argue that regional disparities are deeply shaped by the composition of skills within local labour markets. Some regions host a diverse and complementary portfolio of skills that supports high-value, knowledge-intensive activities, while others are dominated by narrow sets of routine or low-skill occupations. When regions lack a rich and combinable skill base, it becomes more difficult to attract or develop high-productivity industries. In these regions, individuals face fewer opportunities to move into better jobs and higher wages, as a limited skill base constrains the emergence of high-skill occupations and reduces the range of available career ladders. Over time, these factors likely contribute to persistent regional inequalities in opportunities and living standards.

Previous work on economic complexity argues that development depends on the variety and sophistication of the capabilities present in places, originally measured via export baskets and technologies (Hidalgo & Hausmann, 2009). Building on this, new research has started to treat workplace skills as a direct micro-level embodiment of regional capabilities. Buyukyazici et al. (2023) construct regional skill-spaces for Italian regions and show that skill relatedness and skill complexity significantly raise the probability that a region enters new industries and reduce the probability of exit, highlighting that a rich skill base is a key driver of regional diversification and resilience. This project connects to this literature by focusing on skill based regional complexity rather than industry- or product-based complexity.

Studies apply the Economic Complexity algorithm to job-skill networks and derive measures of job fitness and skill complexity based on the rarity and combinatorial richness of required skills (Aufiero et al., 2024). These measures capture how complex occupations rely on many, often rare skills, and they correlate with wages and the position of jobs in the occupational structure. At the regional level, workplace skills as regional capabilities show that skill based relatedness and complexity capture aspects of local capabilities that traditional co-

location measures miss, and that social-cognitive skills in particular are associated with a stronger regional competitive advantage.

This suggests the existence of regional “skill traps”. In many disadvantaged regions, the problem is not only lower average education, but also a narrow and one-sided skill structure, usually dominated by physical or routine work. In such regions it is difficult to attract or grow industries that require complex combinations of high-level (e.g. technical, digital, and social-cognitive) skills. The region can become stuck in a low-skill equilibrium: low complexity jobs dominate, outward mobility to complex occupations is limited, and new industries with higher value added do not emerge. Skill-relatedness studies based on labour flows show that workers tend to move between industries and jobs that are close in their skill requirements, which reinforces path dependence in regional specialisation. (Neffke et al., 2017; Henning et al., 2025)

integrate these papers more

At the same time, the skill structure shapes vulnerability to automation and shocks. Regions specialised in low-complexity, routine-intensive occupations are more exposed to automation risks and to sectoral downturns, while regions with diverse, high-complexity skills have more options to adapt and reallocate workers across activities (Lee et al., 2025). From a policy perspective, this raises the question of whether regional skill complexity can help explain why some regions consistently achieve higher GDP per capita, stronger growth prospects, and more innovation than others, even after accounting for other explanatory factors.

is this really so? manual work is difficult to automate

2. Research question

Building on the problem of regional skill inequality and complexity traps, this project asks whether the structure of skills in local labour markets explain differences in regional development within Europe. Rather than focusing only on how many educated workers a region has, we are interested in how complex the existing skill portfolio is and whether this complexity is associated with higher income levels and stronger innovation performance at the regional level. The main research question is: To what extent does regional skill complexity measured from a region-job-skill multilayer network explain cross-regional differences in economic development and innovation among European regions?

tripartite network?

From this, several more concrete sub-questions follow. Firstly, how is skill complexity distributed across European regions? Using occupations and skill data combined with information on the regional distribution of occupations, we will construct a region-occupation-skill network and apply the „Method of Reflections” by Hidalgo et al. (2007) to derive a skill

how you do this form a tripartite structure?

complexity index for each region. This allows us to map which regions host a rich, diverse and rare skill base, and which are characterised by narrow, low-complexity skill structures.

Secondly, is higher regional skill complexity associated with higher levels of economic development? Here we will test whether regions with more complex skill portfolios have higher GDP per capita. This speaks directly to the idea that regions can be trapped in low-skill equilibria, where limited and routine-intensive skill sets constrain upgrading into high value-added activities.

Lastly, is regional skill complexity positively related to innovation outcomes? We will examine whether regions with higher skill complexity also show higher innovation performance proxied by the number of patent applications relative to the population. This question is motivated by the idea that complex and complementary skills facilitate the combination of knowledge and support the emergence of new technologies and industries.

innovation
is less
motivated

3. Data

Data on occupations and skills will be downloaded from ESCO (European Skills, Competences, Qualifications and Occupations). ESCO is a classification system developed by the European Commission to create a common language for describing work and skills across Europe. It contains standardized occupational titles (ISCO-08 codes at the four-digit level), detailed skill and competence definitions, and explicit links showing which skills are essential or optional for each occupation according to experts (European Commission, 2024).

Since no comprehensive dataset is available to link ESCO's detailed occupation codes to geographic regions, we rely on the Public Use Files from the second cycle of OECD PIAAC. This dataset was collected in 2022-2023 across 31 countries, 22 of which are European, and was originally designed to measure adults' skills across multiple domains. For our purposes, we will only use the background questionnaire, which includes respondents' geographic regions (OECD TL2 level) and their occupations coded in ISCO-08 at the four-digit level. These variables allow us to connect occupations, and through ESCO, their associated skills to specific TL2 regions. During data collection, each participating country used a nationally representative sample of its adult population. However, an important limitation is that it does not contain all countries in Europe.

Furthermore, we will collect region level outcome variables, GDP per capita and the number of patent applications, as well as control variables like population density and educational attainment.

4. Methodology

First, we will construct bipartite edgelists from the ESCO database linking occupations to the skills they require. We then build a second bipartite edgelist connecting regions to occupations, where edge weights correspond to the proportion of workers in each occupation in the region. By merging these two edgelists by their common ISCO-08 codes, we obtain a multilayer, weighted network of regions, occupations, and skills. Because the analysis is based on a multilayer network that links regions to occupations and occupations to skills, we expect the region-skill bipartite network to be densely connected, because of the fact that many occupations are linked to overlapping sets of skills.

To calculate the skill complexity of regions, we will follow the “Method of Reflections” described by Hidalgo et al. (2007). In this framework, the complexity of an entity is derived by combining its diversity (e.g. the number of distinct capabilities or products it has) with the ubiquity of those capabilities (how widespread or rare they are). The result will be a single complexity score that captures how diverse a region’s skill base is and how rare those skills are in other regions.

After constructing the regional skill complexity index, the next step is to investigate its association with key regional outcomes. We will estimate cross-sectional regressions of the following form.

$$Y_r = \alpha + \beta * SkillComplexity_r + \gamma * X_r + \varepsilon_r$$

missing line does not allow for causal language

In the equation above, Y_r denotes GDP per capita or innovation capacity (measured by the number of patents per million inhabitants) in region r and X_r is a vector of control variables. The choice of controls will follow the logic of back-door adjustment, by aiming to block confounding pathways between skill complexity and regional development outcomes. This approach allows us to come as close as possible to identifying a causal relationship by accounting for factors such as population density, education levels, or sectoral structure, that might influence both skill complexity and economic performance.

* Here you assume that skill distribution across regions tells us about their complexity.
what if you measure skill complexity on the occupation level first and then aggregate to regions?

5. Expected outcomes

We expect to find that skill complexity is unevenly distributed across European regions, with some places having broad and rare combinations of skills, while others relying heavily on narrower, routine-intensive skill sets. These descriptive patterns will likely reveal clusters of high- and low-complexity areas.

? what drives the concentration of *
complex skills?

We also anticipate that higher regional skill complexity will be positively associated with higher levels of economic development. Regions with more complex skill portfolios are expected to have higher GDP per capita, reflecting their greater ability to develop high-productivity industries and offer pathways for workers into more complex, well-paid occupations. Finally, we expect that regions with richer and more complementary skill bases will exhibit stronger innovation performance, as the availability of diverse and combinable skills should facilitate knowledge recombination and the emergence of new technological activities. These findings would support the view that skill complexity is a meaningful indicator of a region's development potential.

this could
be tested
with
dynamic
GDP,
innovation
data

6. Impact

This project addresses a gap in the literature on regional development and economic complexity. Most existing work on complexity focuses on industries, products or technologies as carriers of capabilities, using export data or patents to derive complexity indices. Much less is known about how workplace skills, as observed at the level of occupations, aggregate into regional capability structures and how this relates to regional outcomes. By combining ESCO skill information with occupations and regional data from PIAAC, the project develops a skill-based complexity index that is explicitly tied to the composition of local labour markets. This allows us to test whether complexity measures built from job-skill networks add explanatory power beyond standard indicators such as education levels or simple occupational diversity. In this way, the project helps bridge the gap between micro-level skill requirements and macro-level patterns of regional inequality. Besides this, existing studies on skill-relatedness mainly focus on individual countries, such as the U.S. (Alabdulkareem et al., 2018), Italy (Buyukyazici et al., 2024) or Sweden (Henning et al., 2025), and we are not aware of any studies covering the entire European continent.

Beyond its academic contribution, the project has direct societal and policy relevance. It speaks to the problem of persistent regional disparities in opportunities and living standards within Europe, and to the risk that some regions become locked into low-skill equilibria. The

* what should policy do?

↳ FDI

↳ universities

↳ vocational training

main stakeholders include regional and national policymakers, EU cohesion-policy actors, and organisations responsible for skills, training and employment services. The proposed skill complexity index can help identify which TL2 regions lack complex and combinable skills, and which have the potential to diversify into higher value-added activities. Policy makers could use these results to target skill-development programmes, bridge training between related occupations, or develop smart specialisation strategies to the existing skill base rather than only to current industries. For local actors such as regional development agencies, universities or public employment services, the maps of regional skill complexity and relatedness can inform decisions about where to invest in education and training, which types of skills to prioritise, and how to support workers mobility towards more complex jobs without forcing them to leave their region.

At the same time, the project has important limitations and risks. First, the analysis is constrained by data availability: PIAAC only covers a subset of European countries and relies on sample-based information on occupations, and ESCO is an expert-based classification that may not fully reflect real-world skill use in all contexts. The conceptual framework and empirical strategy are portable to other settings: the approach could be replicated for different groups of countries, alternative regional levels (TL3, NUTS3), or other time periods if comparable data on occupations and skills become available. Future research could extend the project by building panel datasets of regional skill complexity, analysing dynamic adjustment to shocks, studying the relationship between skill complexity and automation risk, or zooming in on specific population groups such as young workers or low-skilled adults.

7. References

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